Nile University

School of Information Technology and Computer Science.

CIT 654 - Deep Learning

Submitted to:

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Submitted by:

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Progress report and Contributions

Contributions:

- Using SAM2 for data augmentation to annotate unlabeled dataset.
- Utilizing the temporal (time) features in the U-net. Our proposed approach will be a 3d U-net where the third dimension is the time.
- Analysing data and proposing a new cleansing pipeline (as most approaches use the images raw without processing images). Exploring different scalers and value clippers and testing their effect on the overall segmentation.
- Designing a customized loss function (Dice Loss + Weighted Cross Entropy) to give control over misclassified classes and to reduce perplexity with the background.

Workload distribution:

	Ayman M. Reda	Ahmed A. Selim	Eman K. Ibrahim
CNN	✓		
FCN	✓		
U-Net			✓
Res U-Net			✓
EDA	✓	✓	✓
Custom Loss Function	✓	✓	
Documentation	✓	✓	✓
Local Deployment of SAM	✓	✓	

Current Progress:

- Extensive EDA, resulting in the use of min-max scaler other than Z-scaler and Robust Scaler. Also avoiding 99percentile clipping.
- Implementation of a CNN with a final interpolation (worst model, only made to test a hypothesis of the unsuitability of CNNs and the need for skip-connections).
- Implementation of a FCN to showcase the improvements after adding a dedicated decoder instead of a single interpolator.
- Implementation of a U-net.
- Exercising the effect of using a Cosine Annealing learning scheduler, concluding it yields better results than static learning rates.
- Exercising the effect of using a custom loss function (Dice Loss + Cross Entropy), concluding it yields better results and decreasing background misclassification.

Projected Improvements/Additions:

- Making use of the temporal (time) features.
- Implementing a 3D Res U-Net with our custom loss function and Cosine Annealing scheduler.
- Integrating SAM-based data augmentation in the pipeline.